



Soil type and land use intensity determine the composition of arbuscular mycorrhizal fungal communities

Fritz Oehl^{a,b,*}, Endre Laczko^c, Arno Bogenrieder^d, Karl Stahr^e, Robert Bösch^b, Marcel van der Heijden^a, Ewald Sieverding^f

^aAgroscope Reckenholz-Tänikon Research Station ART, Ecological Farming Systems, Reckenholzstrasse 191, CH-8046 Zürich, Switzerland

^bZürich-Basel Plant Science Center, Institute of Botany, University of Basel, Hebelstrasse 1, CH-4056 Basel, Switzerland

^cFunctional Genomics Center Zurich, University of Zurich, Winterthurerstrasse 180 CH-8057 Zürich, Switzerland

^dInstitute of Biologie II, Research Area Geobotany, Albert-Ludwigs-University of Freiburg, Schänzlestr. 1, D-79104 Freiburg, Germany

^eInstitute of Soil Science, University of Hohenheim, Emil-Wolff-Strasse 27, D-70599 Stuttgart, Germany

^fInstitute of Plant Production and Agroecology in the Tropics and Subtropics, University of Stuttgart Hohenheim, Garbenstr. 13, D-70593 Stuttgart-Hohenheim, Germany

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ABSTRACT

The objective of this study was to test whether soil types can be characterized by their arbuscular mycorrhizal fungal (AMF) communities. To answer this question, a well-defined study area in the temperate climatic zone of Central Europe was chosen with a large spectrum of soils and parent materials. Representative soil samples were taken from three soil types (Cambisol, Fluvisol and Leptosol) at in total 16 sites differing in agricultural land use intensity (9 grasslands and 7 arable lands). AMF spores were isolated and morphologically identified directly from field soils and after reproduction in trap cultures. AMF diversity and community composition strongly depended on soil type and land use intensity, and several AMF species were characteristic for a specific soil type or a specific land use type and hence had a specific niche. In contrast, other AM fungi could be considered as 'generalists' as they were present in each soil type investigated, irrespective of land use intensity. An estimated 53% of the 61 observed AMF species could be classified as 'specialists' as (almost) exclusively found in specific soil types and/or under specific land use intensities; 28% appeared to be 'generalists' and 19% could not be classified. Plant species compositions (either natural or planted) had only a subordinate influence on the AMF communities. In conclusion, land use intensity and soil type strongly affected AMF community composition as well as the presence and prevalence of many AM fungi. Future work should examine how the differences in AMF species compositions affect important ecosystem processes in different soils and to which extent the loss of specific groups of AM fungi affect soil quality.

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1. Introduction

Arbuscular mycorrhizal (AM) fungi are important soil organisms with multiple ecosystem functions since they improve plant growth and health, facilitate nutrient and water uptake (van der Heijden et al., 1998, 2006; Marulanda et al., 2006; Hu and Ruyt, 2007), have positive impacts on soil aggregate stability and water infiltration (Rillig, 2004), and may prevent soil erosion (Schmid et al., 2008). Many AM fungal (AMF) species are ubiquitous since occurring worldwide in quite different terrestrial ecosystems (Öpik et al., 2006), while others appear to be restricted to specific ecosystems, land uses types, vegetations or climates (e.g. Castillo et al., 2006; Oehl and Sieverding, 2004; review in Oehl et al., 2007).

However, little is known about the ecology of AMF in different soils and whether different soil types have specific AMF species and community compositions. Some studies related the occurrence of specific AMF species to soil physical and chemical characteristics such as soil texture, organic matter content and nutrient contents, in particular to the availability of phosphorus (e.g. Uhlmann et al., 2004; Landis et al., 2004; Bashan et al., 2007; Lekberg et al., 2007). However, those studies did not directly compare several different soil types, or they did not differentiate between land use systems (Maia and Trufem, 1990; Land and Schönbeck, 1991). Thus, comprehensive surveys that aim at the comparison of AMF communities in different soils of specific, well-defined climatic or vegetation zones are still lacking. Our objective was therefore to investigate AMF communities of a single (meso-)climate at multiple sites within a well-defined and small as possible area comprising a large spectrum of different soil types, soil parent materials and land use intensities.

* Corresponding author. Tel.: +41 (0)44 377 7321; fax: +41 (0)44 377 7201.
E-mail address: Fritz.Oehl@art.admin.ch (F. Oehl).

For the present study, we have chosen soil types with major distribution characteristics for the temperate climates. We restricted our study area to about 4000 km² in Central Europe where large spectra of soils and land use intensities are close together. In this region, especially in the upper Rhine valley of France, Germany and Switzerland, we had already generated extensive information about AMF communities in two closely related soil types developed on Loess sediments (calcareous Regosols and decarbornated Luvisols; Oehl et al., 2005b, 2009). In these studies we had already included multiple sites with a wide range of agricultural land use intensities (Oehl et al., 2003a, 2004). For the present research questions, soils with a broader range of characteristics were included: siliceous low pH Cambisols developed on nutrient poor, loamy Permian and Triassic sandstones of terrestrial-alluvial origin or on nutrient rich Granite rocks, and on the other extreme, calcareous Leptosols of higher pH developed on marine sediments deposited during Jura age (Herrmann and Stahr, 1998). A low pH Fluvisol with siliceous Granite/Gneiss sediments as soil parent material was also included. The soils included have a sandy-loamy to loamy soil texture but differ – beside in soil pH – largely in mineral (and natural nutrient) composition (e.g. Herrmann and Stahr, 1998; Hädrich and Stahr, 2001).

It was hypothesized that different soil types have characteristic AMF communities and that these are strongly affected by land use intensity. Specifically, we asked in the present study whether (i) AMF community composition depends on soil type, (ii) whether land use intensity (grassland versus arable land) affects AMF community composition across largely differing soil types, and (iii) whether there are AMF species that might be characteristic for specific soils or land use intensities. To answer this questions, field soil samples were taken at the beginning of the vegetative period in spring 2003 and 'AMF trap cultures' were set up to propagate AMF communities under controlled conditions over two complete vegetation periods.

2. Materials and methods

2.1. Study area and locations

The 16 sites selected are all located around the plain of the upper Rhine valley (Oehl et al., 2003a) in small valleys of the bordering mountain ranges of the Black forest (Germany), the Vosges (France) and the Jura Mountains (Switzerland). The (meso-)climate of the region is temperate, with about 700–1000 mm of yearly precipitation at the study sites and an annual average temperature of 8.0–9.5 °C. The soil types at the sites, the types of land use, management practices including type and level of nitrogen and phosphorus fertilization and the geographic position as well as the site abbreviations used in this publication are presented in Table 1. In detail, in seven small valleys of the upper Rhine valley, each a permanent grassland and an arable land were selected (14 sites) at the same elevations (250–450 m a.s.l.) as the Loess soils investigated by Oehl et al. (2003a). Whenever possible, the two land use systems were adjacent to each other (see geographical positions Table 1). Since the Leptosols of the region generally are at slightly higher elevations (here 450–800 m a.s.l.), two additional Cambisol grassland sites (PS-hg and TS-hg) were selected at such higher altitudes (Table 1). It is noteworthy, that in the study area there are no arable lands on Cambisols at these higher elevations. Soils at 5 sites had siliceous sandstone sediments of Permian or lower Triassic origin as parent material and were – according to FAO/ISSS/ISRIC classification (FAO, 2001, 2006) Stagnic Cambisols (having stagnic properties in the subsoil deeper than 50 cm) or Humic Cambisols (carbon content >1% down to 50 cm soil depth) (Table 1). Soils developed on these loamy sandstones are poor in available

nutrients and weatherable minerals (Stahr and Fleck 2004). Seven sites had Granite and Gneiss as parent material and were again Stagnic or Humic Cambisols, or – in close proximity to a small river – Gleyic Fluvisols (Table 1). The soils, which developed on Granite/Gneiss at the foot-hill zone of the Upper Rhine valley, naturally have pH 5.0–5.5 down to 50 cm depth and a significant better nutrient status than the soils which developed on sandstones having pH 4.0–5.0 down to 50 cm depth (Stahr and Jahn, 1989; Herrmann and Stahr, 1998; Stahr and Fleck, 2004). The lasting 4 soils studied were – according to FAO (FAO, 2006) – calcareous, so-called Rendzic Leptosols. They developed on limestone rocks of the Jura Mountains. They have pH 7.0–8.0 (Table 2). Most grasslands were classified as Arrhenathereta with 24–46 different plant species (Table 1). One grassland (JL-g), with 64 different plant species was classified as *Mesobrometum* (Huovinen-Hufschmid and Körner, 1998).

For this study we use three major levels of land use intensity: high, moderate, and low. The high and low land use intensities are subdivided in 'high' and 'very high' and 'low' and 'very low', respectively, resulting in a land use intensity scale from 1 to 5 (Table 1). 'Very high' means maize mono-cropping, and 'high' means narrow (maize-)cereal crop rotation systems, both with > 140 kg N ha⁻¹ input. 'Moderate' means extensive crop rotation systems. 'Low' means grassland systems with more > 2 harvests (cuttings), and 'very low' means grassland with 1–2 harvests or extensive pastures.

2.2. Soil sampling at the field sites

Soil samples were taken from four plots (=4 pseudoreplicates) per field site in early April 2003 when the growing season began in the region. We sampled only once in the season since we had previously found that one sampling date might be sufficient to adequately determine AMF species richness in field soils (Oehl et al., 2005a,b). This is particularly true, when additionally long-term trap cultures (see below) are established to monitor seasonal and successional sporulation pattern of AM fungi (Oehl et al., 2004, 2009). At each of the four plots (with 5 × 20 m size) per field site, six soil-cores were taken from a depth of 0–10 cm using a soil corer of 6 cm diameter. Undisturbed soil crumbs were taken from the 6 soil-cores for setting up AMF trap pot cultures. The rest of each soil core was carefully ground by hand, and the 6 sub-samples per plot were mixed and air-dried. Samples were kept at 4 °C in a refrigerator until further analyses were performed, such as determination of AMF spore density and AMF species richness, and of chemical soil parameters such as pH, organic carbon and available P (Table 2). The chemical parameters were measured in the "Laboratory Dr. Fritz M. Balzer", Wetter-Amönau, Germany, according to standard methods (www.labor-balzer.de). Phosphorus (P) was extracted from the soils with sodium acetate ('P-Morgan') and double lactate ('P-DL'), two wide-spread methods used for estimating plant-available P (Neyroud and Lischer, 2003).

2.3. Trap cultures

For each of the four field plot replicates per site, one AMF 'trap culture' was set up using methods previously described by Oehl et al. (2003a). Briefly, an autoclaved substrate was used consisting of a mixture of Terragreen (American Aluminium oxide, OIL DRY US SPECIAL, Type III R, >0.125 mm, Lobbe Umwelttechnik, Iserlohn, Germany) and Loess (3:1, w/w) and which was placed in a 27 × 17 × 20 cm (L × W × H) plastic box. The AMF inocula consisted of 20 g soil crumbs per trap plant with nine trap plants per pot. The inoculum was positioned beneath the AMF-free trap plants consisting of *Lolium perenne*, *Trifolium pratense* and *Plantago lanceolata*. These trap plants were chosen because they are well

Table 1

Principal agricultural practices, crop rotation, and standing crop at sampling date for field sites.

Soil type on parent material	Land use (Site code)	Farming system	Type and level of fertilization (kg ha ⁻¹)	Land use intensity (scale)	Standing vegetation in April 2003 (number of plant species in grasslands) ^a	Location (m above sea level) and Geographic position (km) ^b
Siliceous soils						
<i>Cambisols on Permian Sandstones (PS)</i>						
Chromi-humic Cambisol	Grassland (PS-hg)	Meadow, conventional, mown 2 × a ⁻¹	Mineral N-P-K fertilizer N (110), P (60)	Very low (1)	Arrhenatheretum (36)	Sankt Peter (780) 427.7 E, 5,319.3 N
Chromi-stagnic Cambisol	Grassland (PS-g)	Pasture, conventional grazed 3 × a ⁻¹	Horse manure and cow slurry N (110), P (50)	Low (2)	Arrhenatheretum (42)	Steinen-Weitenau (420) 408.6 E, 5,281.9 N
Chromi-stagnic Cambisol	Arable land (PS-a)	Maize-wheat crop rotation, conventional	Mineral + slurry N (190), P (50)	High (4)	Bare Soil	Steinen-Weitenau (390) 407.7 E, 5,282.0 N
<i>Cambisols on lower Triassic sandstones (TS)</i>						
Chromi-humic Cambisol	Grassland (TS-g)	Meadow, conventional, mown 2 × a ⁻¹	Mineral N (110), P (50)	Very low (1)	Arrhenatheretum (36)	Sexau-Tennenbach (260) 417.8 E, 5,330.0 N
Chromi-humic Cambisol	Arable land (TS-a)	Maize–maize–wheat crop rotation, conventional	Mineral N (170), P (80)	Highv (4)	Bare Soil	Sexau-Staudenhöfe (250) 418.3 E, 5,331.7 N
<i>Cambisols on Granite rocks (GR) or dilluvial crystalline granite/gneiss sediments (DG)</i>						
Humic Cambisol	Grassland (GR-hg)	Pasture, conventional, grazed 3 × a ⁻¹	Mineral and manure N (100), P (20)	Low (2)	Arrhenatheretum (32)	Gresgen (700) 411.7 E, 5,285.4 N
Humic Cambisol	Grassland (DG-g1)	Meadow, conventional, mown 2 × a ⁻¹	Mineral N (70), P (0)	Very low (1)	Arrhenatheretum (41)	Wintzenheim-La Forge (230) 368.8 E, 5,324.5 N
Humic Cambisol	Arable land (DG-a1)	Maize mono-cropping, conventional	Mineral N (190), P (40)	Very high (5)	Bare Soil	Wintzenheim-La Forge (230) 368.7 E, 5,324.5 N
Stagnic Cambisol	Grassland (DG-g2)	Meadow, conventional, mown 3 × a ⁻¹	Cow manure and slurry N (140), P (60)	Low (2)	Arrhenatheretum (24)	Glottertal (350) 420.7 E, 5,322.1 N
Stagnic Cambisol	Arable land (DG-a2)	Maize-wheat crop rotation, conventional	Mineral + cow manure + slurry N (200), P (70)	High (4)	Bare Soil	Glottertal (350m) 420.5 E, 5,322.1 N
<i>Fluvisols on alluvial crystalline granite/gneiss sediments (AG)</i>						
Gleyic Fluvisol	Grassland (AG-g)	Meadow, conventional, mown 3 × a ⁻¹	Cow manure and slurry N (140), P (70)	Low (2)	Arrhenatheretum (25)	Staufen-Grunern (320) 406.7 E, 5,301.8 N
Gleyic Fluvisol	Arable land (AG-a)	Maize mono-cropping conventional	Mineral N (190), P (60)	Very high (5)	Bare Soil	Staufen-Grunern (325) 405.9 E, 5,302.1 N
Calcareous soils						
<i>Leptosols on Jurassic limestones (JL)</i>						
Rendzic Leptosol	Grassland (JL-hg)	Meadow, bio-organic, mown 2 ×, grazed 1 × a ⁻¹	Sheep manure N (50), P (10)	Very low (1)	Arrhenatheretum (46)	Läufelfingen (610), 413.9 E, 5,251.5 N
Rendzic Leptosol	Arable land (JL-ha)	Grass/clover-cereals-medicinal plants crop rotation, bio-organic	Composted cow manure & slurry N (120), P (50)	Moderate (3)	Grass-clover; 1st year	Läufelfingen (820) 415.3 E, 5,251.0 N
Rendzic Leptosol	Grassland (JL-g)	Pasture (sheep), extensive-conventional	None	Very low (1)	Mesobrometum (64)	Nenzlingen (470) 391.8 E, 5,256.9 N
Rendzic Leptosol	Arable land (JL-a)	Potatoes-cereals-onion crop rotation, bio-organic, no-tillage	Cow manure N (50), P (25)	Moderate (3)	Bare Soil; planted potatoes	Reigoldswil (470) 401.8 E, 5,253.0 N

^a Plant species numbers and plant community structure identified at main flowering of the grasses in springtime (May).^b E, east of Greenwich; N north of the Equator.

known AMF host plants for trap cultures, secondly, they were present at the majority of the grassland sites, and thirdly they represent important functional groups of plants (a grass, a legume and a forb). Four non-mycorrhizal control pots were also included in the study. Irrigation of the plots was by an automated watering system (Tropf-Blumat, Weninger GmbH, 6410 Telfs, Austria). The trap cultures were set up immediately after field sampling. The pots were kept in the greenhouse under ambient natural light and temperature conditions and were maintained over 20 months, from 2003 to the end of 2004. During this time, the above ground part of the trap plants was cut back several times to a level of 3 cm above the ground, generally directly before soil-cores were taken (see below). The control pots remained non-mycorrhizal over 20 months indicating that there was no cross contamination with AM fungi between pots. This long-term culture technique was successfully applied in the past (Oehl et al., 2004, 2005b) and has the advantage that and seasonal and late successional sporulating AMF species can be detected (Oehl et al. 2009).

2.4. Sampling in the trap cultures

For the analyses of AMF spores, two soil-cores were taken at the same locations within each trap culture box at 2, 4, 6, 8 and 14,16, 18 and 20 months after planting. The core volume was 15 cm³ and the sampling depth 10 cm. After sampling the resulting holes in the boxes were immediately refilled with the sterile soil substrate used for the trap pot culturing. The initial rates of spore formation and mycorrhizal root colonization were determined to investigate if both parameters are affected by soil type or land use intensity. Spore formation was determined by extracting and counting the spores formed two and four months after initiation of the pot culturing as described in Oehl et al. (2003a) while mycorrhizal root colonization was determined as described by Brundrett et al. (1994) after 2 months only. Since we observed that extensive analyses of the field samples gave broader information on AMF species compositions and diversities than extensive analyses of the trap cultures we hereafter focus mainly on the field soil analyses.

Table 2

Chemical soil parameters at field sites differing in soil type and agricultural land use.

Soil type and parent material	Land use (site code)	pH		Organic carbon (mg kg ⁻¹)	Available phosphorus (mg kg ⁻¹)	
		H ₂ O	KCl		Sodium acetate ^a	Double lactate ^a
Siliceous soils						
Cambisols on Permian Sandstones (PS)	Grassland (hg)	5.3	4.2	52.2	5.2	30.6
	Grassland (g)	5.6	4.7	38.3	4.4	24.9
	Arable land (a)	5.6	4.5	18.0	6.1	34.5
Cambisols on lower Triassic Sandstones (TS)	Grassland (g)	6.0	5.1	30.2	9.6	64.2
	Arable land (a)	6.1	5.2	14.5	27.1	203
	Cambisols on Granite rocks (GR) or dilluvial crystalline Granite/Gneiss sediments (DG)	Grassland (hg)	5.6	4.4	47.6	5.7
Grassland (g1)		5.7	4.8	38.2	7.9	41.9
Arable land (a1)		6.3	5.4	20.2	10.0	46.7
Grassland (g2)		6.2	5.3	48.7	14.0	55.0
Arable land (a2)		6.2	5.5	32.5	9.2	62.0
Fluvisols on alluvial crystalline Granite/Gneiss sediments (AG)	Grassland (g)	5.6	4.9	52.2	19.2	49.3
	Arable land (a)	5.4	4.2	21.5	15.7	31.4
Calcareous soils						
Leptosols on Jurassic limestones (JL)	Grassland (hg)	7.6	6.9	47.6	23.6 ^b	95.2 ^b
	Arable land (ha)	7.8	7.1	38.3	6.1	13.1
	Grassland (g)	7.7	6.9	44.1	2.2	5.7
	Arable land (a)	7.7	7.2	42.9	2.6	6.5

^a Soil extractant; see [Materials and methods](#).^b Increased available soil phosphorus values due to high-input fertilization with conventional chicken manure before 1995.

2.5. AMF spore isolation and identification

The AMF spores were isolated from the original soil samples from each of the four replicates taken at the 16 field sites, and identified and counted separately for each replicate. AMF spores were extracted from the field samples and from the trap cultures by wet sieving and sucrose density gradient centrifugation (Oehl et al., 2003a). Spores were counted under a dissecting microscope using up to 90-fold magnification. For identification of AMF species, spores (about 50–70% of the total number) were picked under the dissecting microscope and mounted in polyvinyl alcohol–lactic acid–glycerine (PVLG) or PVLG mixed 1:1 (v/v) with Melzer's reagent (Brundrett et al., 1994) for permanent slides. The spores were then examined using a compound microscope at up to 400-fold magnification. The AMF species identification was based on current species descriptions and identification manuals (Schenck and Pérez, 1990; International Culture Collection of (Vesicular) Arbuscular Mycorrhizal Fungi (<http://invam.caf.wvu.edu/fungi/taxonomy/speciesID.htm>)) and on own type specimen analyses of almost all AMF species (>95%) described. In this study, an AMF “species” is either a clearly identified species based on its spore morphology, or an as yet unknown species to us. The nomenclature “species-group” comprises more than one species due to the fact that the morphological features of some spores of that group were not clear enough to place all spores to one single defined AMF species. We deposited specimens of almost all species that were detected in this study at the mycological herbarium Z + ZT in Zurich.

2.6. Statistics and multivariate data analysis

Data for AMF spore density, AMF species and genus richness or diversity (Shannon–Weaver index) in the field soil samples, and AMF root colonization and new spore formation in the trap culture systems were reported as mean and standard deviation of four field plot or trap culture replicates. For all these parameters, the significance of differences between field sites were tested by using Fisher's least significant difference (LSD) at $p < 0.05$ after test for normality and performance of one-way ANOVA. Initially, we had inspected the distribution of all data graphically in respect of the factors site, land use intensity and soil type (boxplots and Cleveland dot plots, not presented) and decided to perform, as planned, only 1-way ANOVA for differences by sites since 2-way ANOVA would

have been to be performed on our unbalanced data set characterized by the lack of a proof of homoscedity (i.e. low sample numbers, unequal variance, groups without observations, normality test not applicable for all groupings). Moreover, we found that other factors like altitude, even when over a few hundred meters only, might be affecting the results, too. Instead, we decided to perform two different multivariate analyses (Hierarchical Cluster Analysis HCA and Canonical Correspondence Analysis CCA) in order to estimate and to visualize the impact of land use and soil type on the AMF communities.

Dependency of selected AMF parameters (e.g. spore density of specific species) on selected soil parameters or plant species richness in grasslands was demonstrated by using linear regression analyses testing the mean value of each soil parameter and total plant species richness per grassland site, respectively, against the total AMF species richness found per site.

The Shannon–Weaver (H') index is generally calculated as a measure of AMF species diversity, combining two components of diversity, richness and evenness. Here, the index was also applied on the genus level for describing AMF genus diversity. To calculate the species richness and the Shannon–Weaver index on the species level, the two AMF species grouped into one “species-group” were counted as one single species.

To analyze the relatedness of sampling sites in respect of AMF species composition (AMF communities), a hierarchical cluster analysis (HCA) using Ward's minimum variance method (e.g. Legendre and Legendre, 1998) was performed whereby we used the χ^2 distances matrix of the sampling sites, calculated from AMF spore densities. Relations between AMF species composition and the soil parameters determined were identified by canonical correspondence analysis (CCA) according to ter Braak (1986). AMF species that were found solely at one of the sites or – in the whole study area – only in very infrequent numbers (≤ 5 spores) were not considered for in this analysis. For HCL and CCA the software package ADE-4 was used (Thioulouse et al., 1997).

3. Results

3.1. AMF species found at field sites

In total 58 species could be distinguished from the field samples across all sites (Table 3 and 4). In one case, where two species could

<i>Entrophospora infrequens</i>		0.2 (1)		0.2 (1)		0.2 (3)	0.5 (1)	0.1 (1)		0.3 (1)		0.6 (7)	1.7 (6)	0.8 (9)	0.3 (1)	G
Acaulosporaceae																
<i>Acaulospora paulinae</i>	5.4 (17)	17.6 (110)	2.4 (4)	18.0 (83)	6.0 (10)	17.6 (46)	18.3 (304)		4.9 (41)	28.0 (88)	6.0 (20)					S & L
<i>Ac. laevis</i>		5.2 (33)	0.6 (1)	2.8 (13)	3.0 (5)	1.9 (5)	1.3 (21)	1.4 (3)	0.1 (1)	1.9 (6)				0.1 (1)		S & L
<i>Ac. cavernata</i>		0.6 (4)	2.4 (4)	0.9 (4)					0.9 (8)			0.3 (1)	2.2 (3)			S & L
<i>Ac. longula</i>	0.9 (3)	2.4 (15)		0.9 (15)		2.7 (7)	0.7 (12)	0.5 (1)			0.3 (1)			0.2 (2)		S & L
<i>Ac. scrobiculata</i>		0.5 (3)	0.6 (1)			0.4 (1)										S
<i>Ac. thomii</i>		0.5 (3)				0.8 (2)	0.5 (8)							0.2 (2)		S
<i>Ac. sp. BR19</i>		0.3 (3)			3.6 (6)		1.5 (25)			1.9 (6)						S
<i>Ac. polonica</i>			0.6 (1)					0.6 (5)								N
<i>Ac. sp. BR20ⁱ</i>		0.3 (2)														N
<i>Ac. myriocarpa</i>	0.6 (2)			0.4 (2)		0.4 (1)					0.3 (1)					S & L
<i>Kuklospora colombiana</i>		0.9 (6)		0.2 (1)		0.4 (1)										S & L
Scutellosporaceae																
<i>Scutellospora calospora</i>	4.4 (14)	1.1 (7)	1.8 (3)	7.6 (35)		5.7 (15)	1.5 (25)	0.5 (1)	1.3 (11)	0.3 (1)	1.8 (6)	0.7 (1)	0.2 (3)	3.2 (11)	8.5 (25)	G (L)
<i>Sc. dipurpurens</i>	2.8 (9)					11.9 (31)										S & L
<i>Sc. arenicola</i>									0.2 (2)							N
Racocetraceae																
<i>Cetraspora armeniaca</i>	1.6 (5)	0.9 (6)		3.7 (17)			1.6 (27)		1.4 (12)		2.7 (9)					S & L
<i>Ce. pellucida</i>		0.3 (2)		5.4 (25)			2.8 (46)		1.4 (12)		3.0 (10)			0.4 (4)		S & L
<i>Racocetra castanea</i>								0.9 (2)			1.2 (4)					N
Gigasporaceae																
<i>Gigaspora margarita</i>	0.9 (3)		0.6 (1)	0.2 (1)		1.5 (4)	0.7 (12)				0.6 (2)	0.7 (1)				S & (L)

^a See Table 1.

^b Estimates for AMF species specificity/preference for specific soil types (S) and/or land use intensity (L), or for non-specificity (G), as species were found to have specific niches (being so-called 'specialist' AM fungi), or were ubiquitous 'generalist' AM fungi; rarely detected species were not estimated (N)

^c Comprises *Gl. albidum* and *Paraglomus occultum*.

^d Resembles *Gl. melanosporum*.

^e Resembles *Gl. coremioides*.

^f Resembles *Gl. epigaeum*.

^g Resembles *Gl. tortuosum*.

^h Resembles *Am. appendicula*.

ⁱ Resembles *A. rehmi*.

Gl. aureum and *Gl. badium* were recently described from the study area (Oehl et al., 2003b, 2005a), formerly presented as *G. sp.* strain BR2 and *G. sp.* strain BR4 in Oehl et al. (2003a, 2004, 2005a), respectively.

Ambispora fennica (Walker, 2008) was called *Acaulospora sp.* BR13 in Oehl et al. (2004).

Pacispora dominikii is the former *Glomus dominikii* (Oehl and Sieverding, 2004).

Kuklospora colombiana is the former *Entrophospora colombiana* (Sieverding and Oehl, 2006).

The families Scutellosporaceae and Racocetraceae were described recently (Oehl et al., 2008).

Table 4

AMF species richness found at field sites and in trap cultures and Shannon–Weaver species and genus diversity index calculated from the number of species respective genera identified.

	Siliceous soils											Calcareous soils				LSD	
Soil type	Cambisols										Fluvisols		Leptosols				
Geology	Permian Sandstones (PS)			Triassic Sandstones (TS)		Granite rocks (GR) or Dilluvial Granite/Gneiss sediments (DG)					Alluvial Granite/Gneiss sediments (AG)		Jurassic Limestones (JL)				
Land use type	Grassland	Grassland	Arable	Grassland	Arable	Grassland	Grassland	Arable	Grassland	Arable	Grassland	Arable	Grassland	Arable	Grassland		Arable
Site code	PS-hg	PS-g	PS-a	TS-g	TS-a	GR-hg	DG-g1	DG-a1	DG-g2	DG-a2	AG-g	AG-a	JL-hg	JL-ha	JL-g	JL-a	
Field site analyses																	
AMF species richness																	
At field sites ^a	17.5 d	21.8 ab	12.0 f	20.0 bcd	9.5 g	16.8 d	23.3 a	14.3 e	19.5 c	12.0 f	18.5 cd	10.0 g	18.5 d	13.0 ef	20.3 bc	11.3 fg	1.92
Total at field sites ^b	25	34	19	31	15	25	32	21	25	17	30	16	23	16	31	15	
Shannon–Weaver species diversity index ^a	2.34 bcd	2.52 ab	2.01 f	2.54 a	1.66 g	2.42 abc	2.38 abcd	2.31 cd	2.21 de	2.01 f	2.48 abc	1.58 g	2.31 cd	2.03 ef	2.29 d	2.02 f	0.19
Shannon–Weaver genus diversity index ^a	0.79 b	0.80 b	0.47 cde	0.94 ab	0.38 def	1.11 a	0.91 b	0.33 efg	0.57 cd	0.64 c	0.66 c	0.14 gh	0.11 h	0.23 fgh	0.10 h	0.29 efgh	0.20
Trap culture analyses																	
AMF species richness in trap cultures after 20 months	21	21	16	16	20	18	26	16	19	17	20	13	20	19	21	21	
Summary of field and trap culture analyses																	
Species richness from field sites and trap cultures	32	39	23	35	23	32	38	25	32	21	33	21	27	20	33	24	
Ratio of species found in samples from trap cultures/field sites	0.84	0.62	0.84	0.52	1.33	0.72	0.81	0.76	0.77	1.00	0.67	0.81	0.87	1.19	0.68	1.40	

^a Mean for four replicate plots at field site. Nonsignificant differences between sites are shown by identical letters and were determined with Fischer's Least Significant Difference (LSD) test at the 5% level after a one-way ANOVA.

^b Sum of four field plots per field site.

not unequivocally be distinguished, a species group was defined ('*Glomus occultum*' group; Table 3). Eleven species could not clearly be named at the species level (e.g. *Acaulospora* sp. BR19, *Ambispora* sp. BR10 resembling *Ambispora appendicula*). A few of them are assumed to be as yet un-described species. The majority of the isolated species were of the genus *Glomus* (Gl., Table 3). Two of them (*Glomus sinuosum* and *Glomus rubiforme*) had formerly been assigned to *Sclerocystis*. Ten *Acaulospora* (Ac.), two *Archaeospora* (Ar.), two *Ambispora* (Am.; Spain et al., 2006; Walker, 2008), and each one *Entrophospora* (En.), *Kuklospora* (Ku.), *Paraglomus* (Par.) and *Pacispora* (Pac.) were found. Three species were within the genus *Scutellospora* (Sc.), two species were within *Cetraspora* (Ce.) each one within *Racocetra* (*Racocetra castanea*) and *Gigaspora* (*Gigaspora margarita*).

3.2. AMF spore density

The AMF spore densities differed among the various soil types and depended on land use. Grasslands had regularly higher densities than arable lands. Leptosols soils contained higher spore numbers compared to Cambisols or Fluvisols. This applied for grasslands as well as for arable lands (Fig. 1).

3.3. AMF species composition

Irrespective of soil type, grasslands generally contained more AMF species and genera compared to arable soils. AMF species diversities, expressed by the Shannon–Weaver index, were generally also lower in the arable lands (Table 4). In contrast, species richness did not vary significantly among soil types and ranged between 25 and 34 species in Cambisol/Fluvisol and Leptosol grassland (Table 4) and between 15 and 21 species in arable lands (Table 4). AMF species diversities were also similar in the grasslands of all the soil types investigated (Table 4), with slightly lower values in the Leptosols.

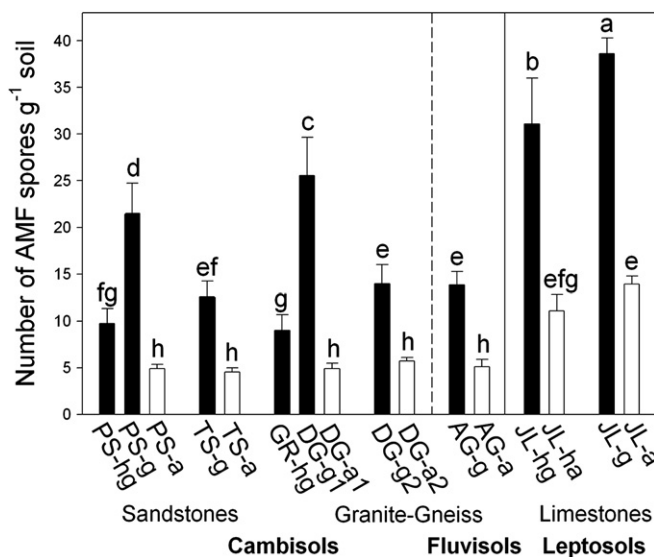


Fig. 1. AMF spore density at the field sites with different geology, soil types and cultivation practices applied (see Table 1). Data are reported as averages and standard deviations for four replicate plots per site. Nonsignificant differences between sites are indicated by identical letters above the bars and were assigned by comparison with (Fisher's) Least Significant Difference at the 5% level after one-way ANOVA. Black bars represent grassland sites, while white bars represent arable lands. For site codes see Table 1.

In the Leptosols, 95–99% of all the spores found belonged to the genus *Glomus*, whereas in the Cambisols only 55–80% of the spores belonged to *Glomus* (Tables 3 and 4; Fig. 2). All species of *Acaulospora*, *Kuklospora*, *Scutellospora*, *Racocetra*, *Cetraspora* and *Gigaspora* had significantly higher spore densities and species richness in the Cambisols compared to the Leptosols (Table 3). In the Leptosols, only three *Acaulospora* spp. and one *Scutellospora* sp. were found, while from the Cambisols 10 and 4 were recovered, respectively. *Ambispora fennica*, *Kuklospora colombiana*, *Cetraspora armeniacae* and *Gi. margarita* were only found in the Cambisols (Table 3). Interestingly, the number of sporocarpic *Glomus* species was highest in the Leptosols (Table 3), and also *Pacispora dominikii* spores were – even though generally in low numbers – regularly found in the Leptosols while they were generally absent in the Cambisols and never found with soil pH < 6.0 (Table 3). On the genus level, though, AMF spore populations (Fig. 2) were clearly more diverse in the Cambisols and Fluvisols than in the Leptosols, and both AMF genus richness (Fig. 3) and diversity (Shannon–Weaver index, Table 4) were higher for Cambisol than for Leptosol sites, except for the two siliceous soils subjected to the highest land use intensity of all soils studied (DG-a1, AG-a).

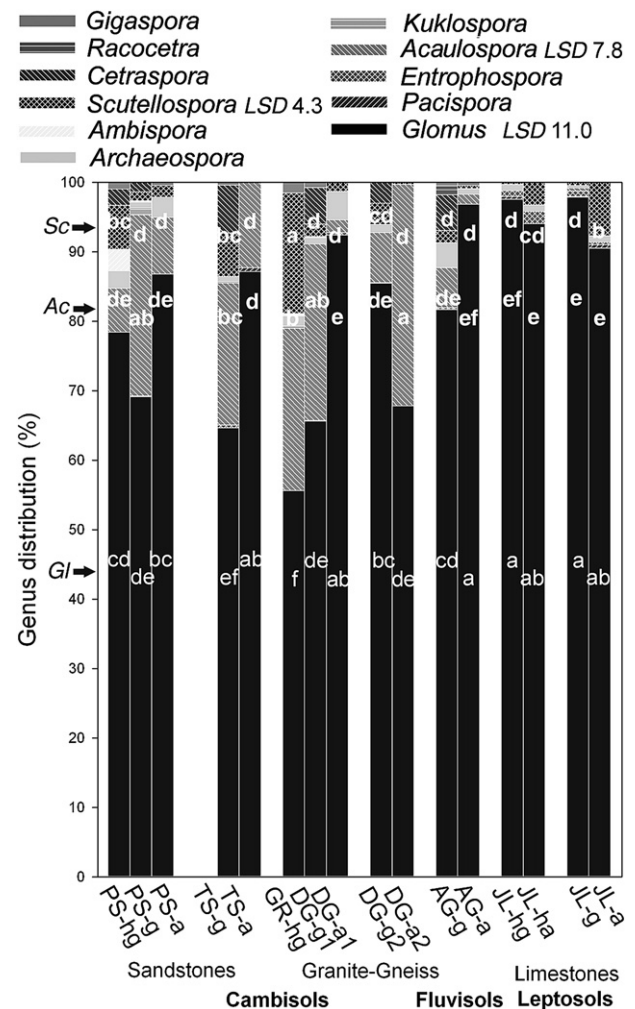


Fig. 2. AM fungal spore compositions analyzed on the genus level at the field sites with different geology, soil types and cultivation practices applied (see Table 1). Data are reported as mean of four replicate plots per site. For the three most frequent AMF genera *Glomus* (Gl), *Acaulospora* (Ac) and *Scutellospora* (Sc), nonsignificant differences between sites are indicated by identical letters within a row, assigned by comparison with (Fisher's) Least Significant Difference (LSD) at the 5% level after one-way ANOVA. Each bar represents a study site (see Table 1).

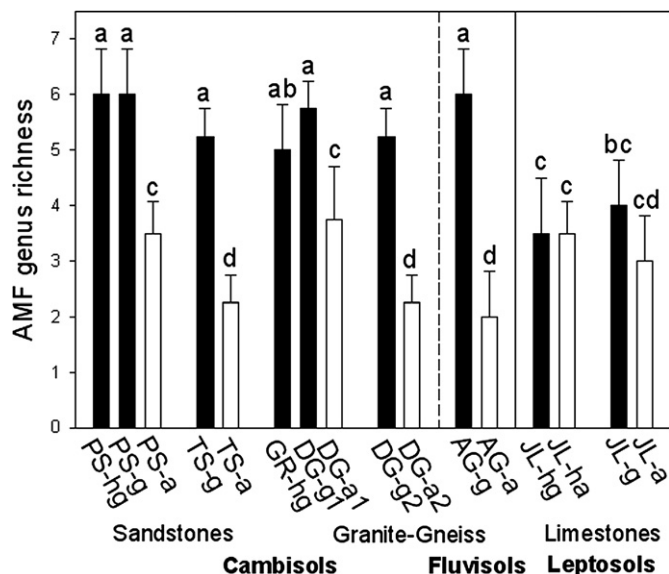


Fig. 3. AM fungal genus richness at the field sites with different geology, soil types and cultivation practices applied (see Table 1). Data are reported as averages and standard deviations for four replicate plots per site. Nonsignificant differences between sites are indicated by identical letters above the bars and were assigned by comparison with (Fisher's) Least Significant Difference at the 5% level after one-way ANOVA. Black bars represent grassland sites, while white bars represent arable lands. For site codes see Table 1.

Several species occurred at all field sites regardless of the soil type and agricultural land use system (Tables 3 and 4). Such AMF 'generalists' were *Glomus diaphanum*, *Glomus fasciculatum*, *Glomus intraradices*, *Glomus etunicatum*, 'Gl. occultum' group and, surprisingly, also *Archaeospora trappei* and *Scutellospora calospora* (Fig. 4).

Other species apparently were more or less restricted to specific situations. Some species were more abundant in arable lands when compared to grasslands. These were for example *Glomus mosseae* in the Leptosols and in Cambisols and Fluvisols developed on Granite/Gneiss. Moreover, *Glomus caledonium* occurred almost exclusively in the Cambisols (Fig. 5). Other species like *Glomus aureum* and *Glomus macrocarpum* were more numerous in the grasslands than in the arable lands (Fig. 5). Concomitant increased spore densities of *Gl. macrocarpum* and *Acaulospora paulinae* were most characteristic for the Cambisol grasslands, and especially *Ce. armeniacae*, *Cetranspora pellucida*, *Gl. margarita* and other *Acaulospora* species were most abundant at such sites (Fig. 5). Increased densities of *Glomus constrictum* together with several sporocarpic *Glomus* spp. (e.g. *Gl. sinuosum* and *Gl. rubiforme*) were most typical for the Leptosol grasslands (Fig. 5 and Table 3). Dominance of spores of *Gl. constrictum* and the occurrence of sporocarpic *Glomus badium* was most obvious at the no-tilled cultivated Leptosol site JL-a (Table 3). Remarkably, especially *Ac. paulinae* sometimes occurred in higher numbers in crop rotation systems than in adjacent grasslands, but it was never recovered from maize mono-cropping systems (DG-a1, AG-a).

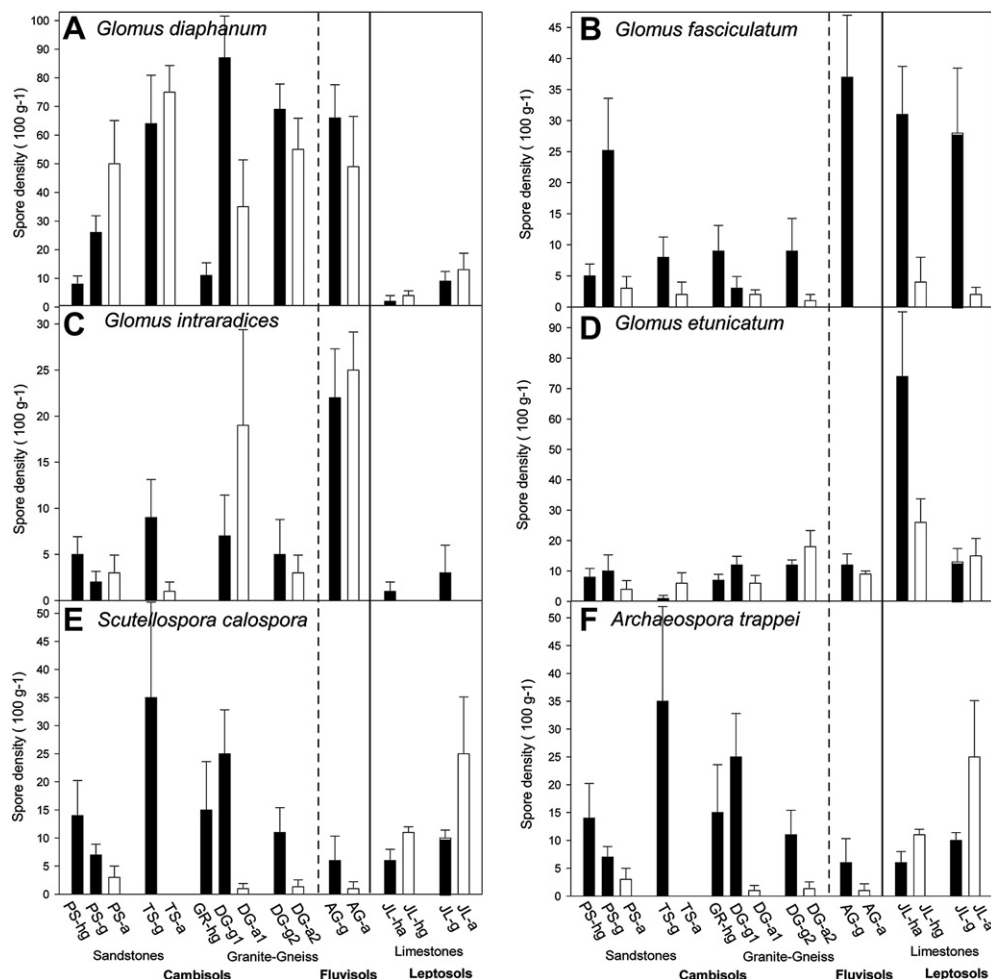


Fig. 4. Spore density (in field samples) of selected AMF 'generalist' species, not or only minor affected by land use system or soil type. Data are reported as averages and standard errors for four replicate plots per site. Black bars represent grassland sites, while white bars represent arable lands. For site codes see Table 1.

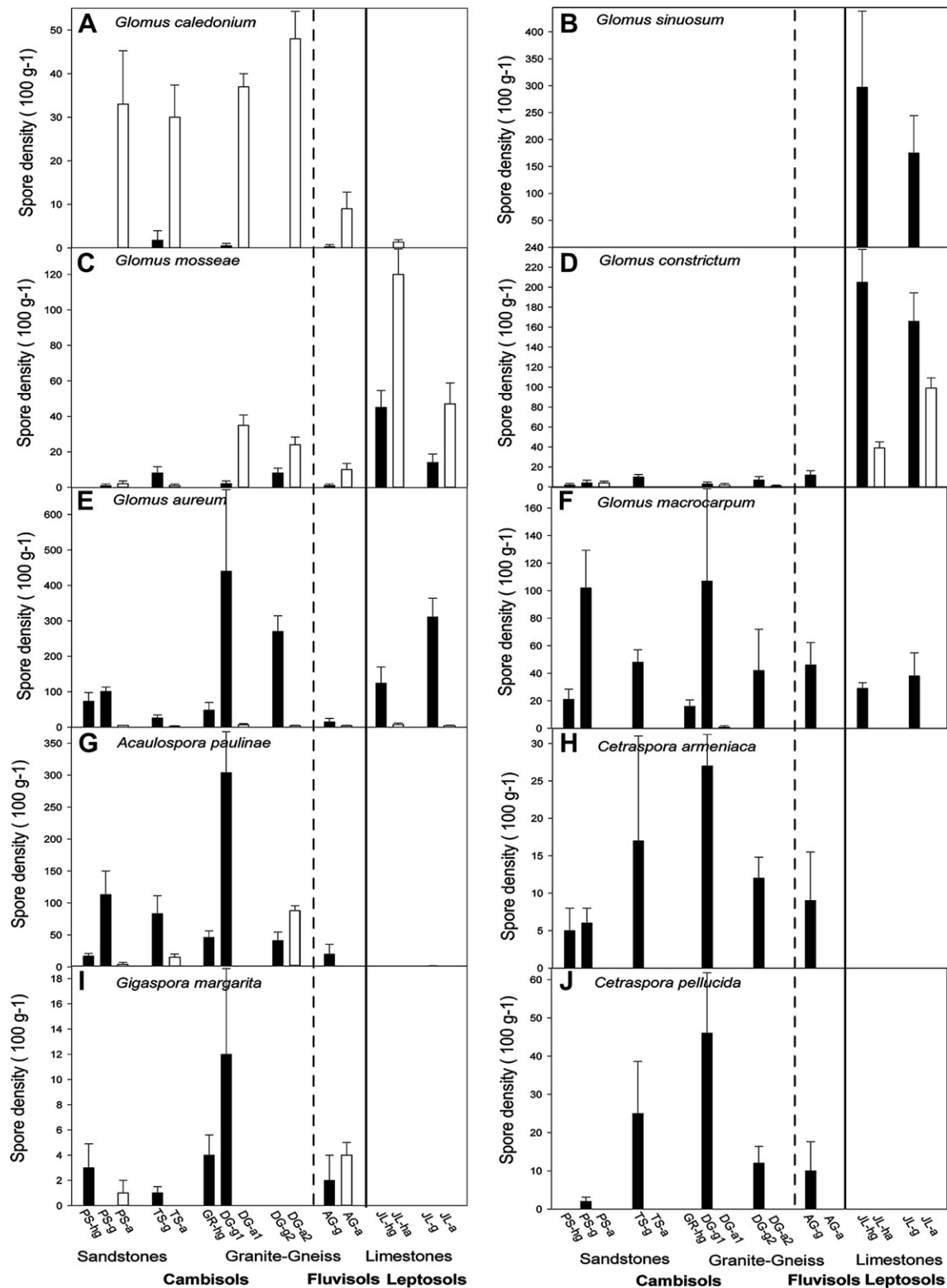


Fig. 5. Spore density (in field samples) of selected AMF 'specialist' species, affected either by land use intensity and/or by soil type. Data are reported as averages and standard errors for four replicate plots per site. Black bars represent grassland sites, while white bars represent arable lands. For site codes see Table 1.

3.4. Cluster analyses

Hierarchical cluster analyses (HCA) showed that land use intensity was the dominating factor which separated the similarities in AMF species compositions between the field sites (Fig. 6).

At the lower levels of land use intensity, the AMF communities of the four (calcareous) Leptosols separated clearly from the siliceous Cambisols/Fluvisols. The arable lands and grasslands of the Leptosols separated on a third level. On a third level also the two grasslands of the Cambisols, located on an about 300–500 m

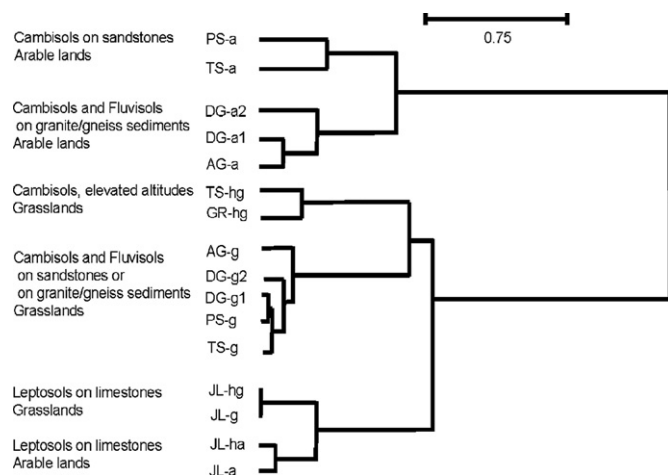


Fig. 6. Hierarchical cluster analysis (HCA) using the χ^2 distance matrix of the field sites (see Table 1) based on the χ^2 distance (measure of scale for 0.75 units) calculated from the site specific AMF species composition, and the Ward's minimum variance method. Note the close similarity of the two calcareous arable lands, of the two calcareous grasslands, and of the five arable lands on siliceous bedrocks. Also the seven siliceous grasslands group together, but two of them, which are located at about 300–500 m higher altitude (see Table 1), form a subcluster distinct from the other five siliceous grasslands.

higher altitude (PS-hg and DG-hg), separated from the Cambisol grasslands grown at the foot-hill zone (e.g. PS-g, DG-g1 and DG-g2). Within the high intensive land use systems, the AMF communities were different for Cambisols developed on nutrient poor sandstones compared to those Cambisols/Fluvisols developed on Granite/Gneiss rocks and sediments.

Canonical correspondence analysis (CCA) revealed clear relationships between AMF species communities, the soil types and land use intensities (Fig. 7). Four major groups of AMF community clusters were formed. Species of the first cluster (e.g. *Gl. caledonum* and *Gl. intraradices*) grouped with the sites of highest land use intensities (4 and 5); i.e. with the Cambisol arable lands of maize mono-cropping and narrow crop rotation systems. Almost all species of Acaulosporaceae and Gigasporineae (e.g. *Gi. margarita*, *Ce. armeniacae* and *Sc. calospora*), but only a few *Glomus* spp. (e.g. *Gl. invermaium*, *Gl. macrocarpum* and *Glomus hoi*) grouped with the 'low' and 'very low' intensity (intensity 1 and 2) grassland Cambisols. Many *Glomus* species (e.g. *Gl. constrictum* and *Gl. sinuosum*) and *Entrophospora infrequens* grouped with the low intensity (1) grassland Leptosols. The arable land Leptosols with wide crop rotation and moderate to low intensity (3) clustered only with a few species: *Glomus geosporum* and *P. dominikii*. Some species fell out of the clusters (e.g. *Gl. etunicatum*) or are positioned in between two associated clusters (e.g. *Gl. mosseae*, or *Gl. aureum* and *Gl. badium*), and thus were actually not associated with a soil type or specific land use.

The soil nutrient status indicators 'P-NaAC' (=P-Morgan) and P-bilactate (=P-DL) did not contribute to the grouping of AMF species or the separation of the sites and were not correlated with the distribution of any AMF species. Only, a slightly positive correlation of P-NaAC with the '*Gl. occultum*' group and a slightly negative correlation with *Sc. calospora* was found.

3.5. Relationship of selected chemical soil parameters and plant species richness to AMF genus richness and relative densities of *Glomus* spores at site

AMF genus richness (Fig. 3) was negatively and the relative density of *Glomus* spores was positively related to soil pH (Table 2)

($r^2 = 0.25$ and 0.27 , respectively). When the linear regressions were calculated only for the grassland sites, even stronger relationships between soil pH and AMF genus richness and relative density of *Glomus* spores were revealed ($r^2 = 0.68$ and 0.57 , respectively), while no relationship was found when analyses were performed only for cultivated sites ($r^2 = 0.06$ and 0.02 , respectively). There were no relationships between organic carbon content respective P availability and AMF genus richness or spore populations. Interestingly, there was also no significant relationship between AMF genus richness (Fig. 3) and plant species richness (Table 1) at the nine grassland sites ($r^2 = 0.14$).

3.6. Initial AMF root colonization and spore formation in the trap cultures

The mycorrhizal root colonization in the trap cultures after 2 months was higher in the cultures of the grasslands than of the arable lands (Fig. 8a). Remarkably, one exception was found for the cultivated Leptosol site JL-a under no-till where root colonization was on a similar level as in the Leptosol grasslands.

Spore formation after 4 months trap culturing was higher in the arable lands than in the adjacent grasslands (Fig. 8b). After 20 months, 40 AMF species had sporulated in the trap cultures (data not shown) of which three species had not been detected from the field samples. These fungi, *Glomus lamellosum*, *Glomus* sp. BR11 and *Glomus coronatum*, appeared across all soil types and land use intensities. AMF species richness was generally higher in the grassland systems than in the arable lands, but no clear pattern was found between soil types (Table 4).

3.7. Total AMF species richness found at field sites and in trap cultures

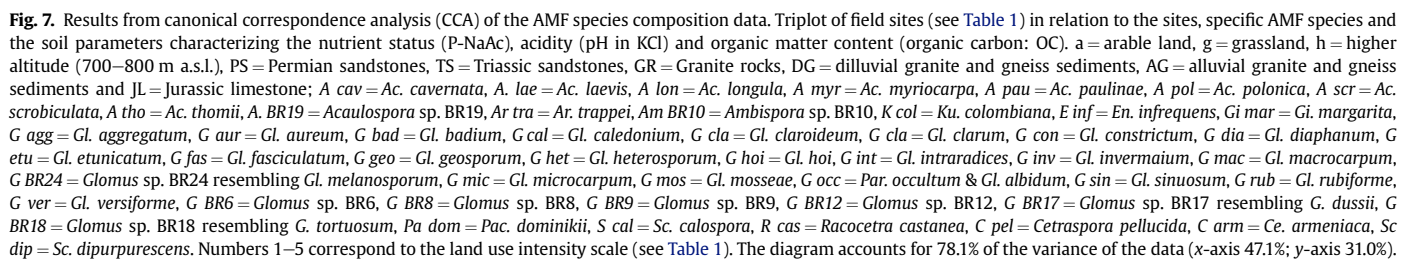
In total, 61 AMF species were identified either directly in the field soils (58 species) or in trap cultures (3 additional species among 40 species detected). Total AMF species richness was slightly higher in the Cambisols/Fluvisol grasslands (32–39 species) than in the Leptosol grasslands (27 and 33 species; Table 4). Much lower species richness was found in the arable lands ranging between 20 and 25 species in the three soil types investigated (Table 4). An estimated 53% of the in total 61 observed AMF species could be classified as 'specialist' fungi as (almost) exclusively found in specific soil types (15%) or land use intensities (15%), or in specific soil types under specific land use (23%) (Table 3); 28% appeared to be AMF 'generalists' since they occurred in all soil types and land use intensities, and 19% could not be classified, since these AMF fungi were only rarely detected in the study area (Table 3).

3.8. Correlation between plant species richness and AMF species richness

There was no correlation between plant species richness (Table 1) and AMF species richness (Table 4), neither when calculated for AMF species richness found in field samples alone nor for the AMF species richness found through combined field and trap culture analyses ($r^2 = 0.04$ and 0.00 , respectively).

4. Discussion

The results of this investigation show that land use intensity and soil type are major determinants of the composition and species richness of AMF communities in Central European soils. Our earlier findings from the same study area, that increasing land use intensity negatively affects the diversity of the AMF communities on (calcareous) Regosols and (decarbonated) Luvisols developed on



Our analyses showed that Leptosols and Cambisols have characteristic AMF species compositions: their spore populations largely differed, and specific species and species combinations were identified that stand for the differences in the community structure between the two soil types. Most noteworthy, the genus richness and genus diversity (Shannon–Weaver index) were significantly higher in the Cambisols than in the Leptosols. The findings correspond well with results of [Oehl et al. \(2003a, 2005b\)](#) and [Sýkorová et al. \(2007a\)](#) obtained from decarbonated Luvisols, calcareous Regosols, and a calcareous Leptosol applying morphological and molecular analyses, respectively.

Some species had their preferred niches even if they appear to belong rather to the ubiquitous 'generalists'. The dominance of *Gl. constrictum* and the concomitant occurrence of sporocarpic *Gl. badium* were most obvious in the calcareous Leptosols that were

not or only minor exposed to mechanical disturbance (Table 3). Earlier studies found *Gl. constrictum* in arable lands with wider crop rotations and in grasslands (Błaszkowski, 1993), above all in soils of soil pH (>6.0) (Oehl et al., 2003a, 2009), so that we now conclude that *Gl. constrictum* is a character species of such agriculturally used soils in Central Europe. On the other hand, the increased presence of several sporocarpic species was typical for calcareous grasslands in the study area, whereas *Gl. geosporum* was abundant both in arable lands and in grasslands of higher pH soils. Our finding is in accordance with Landwehr et al. (2002) and Hildebrandt et al. (2001) who found *Gl. geosporum* dominantly in soils of pH 8.5 in Germany. In calcareous arable lands with higher land use intensity, species like *Gl. mosseae* may increasingly replace species like *Gl. geosporum* and *Gl. constrictum* (Fig. 5, Table 3).

Independent of the soil type, other species like *Gl. aureum* and *Gl. macrocarpum* were much more abundant in grasslands than in arable lands where especially *Gl. macrocarpum* was generally absent. Remarkably, it took 16–20 months for *Gl. macrocarpum* to sporulate in the trap cultures from the grasslands, which may explain why it did not occur in the cultivated soils, while *Gl. aureum* generally sporulates (slightly) faster depending on the year (data not shown for this study; Oehl et al. 2009). The two species, and especially *Gl. macrocarpum*, are thus character species for such grassland sites (e.g. Oehl et al., 2003b, 2005a).

Soil types differ in soil chemical and physical characteristics, and the question is whether the occurrence of specific AMF species was determined by those properties. Earlier work showed that pH (e.g. Clark, 1997; Giovannetti, 2000), organic matter content (e.g. Sieverding, 1989) and soil P availability (e.g. Oehl et al., 2004) influence the composition of AMF communities and the abundance/occurrence of specific AMF species. Our data suggest that soil pH is the parameter that most influences the AMF species composition in soils (Figs. 3 and 8), especially under lower land use intensity. Organic carbon and available P were often used as parameters to which the AMF spore densities and species richness were correlated (e.g. Sieverding, 1991; Uhlmann et al., 2004; Landis et al., 2004; Bashan et al., 2007), and in some studies they were correlated with species diversity (e.g. Tchabi et al., 2008; Oehl et al., 2004). In our study, these parameters did not affect the AMF spore populations, but at least the organic carbon contents appeared to be correlated with a shift in the AMF community (Fig. 7) and with a higher infectivity in the trap cultures (Fig. 8). However, this is likely an indirect relationship since for a given soil type, organic carbon is highly dependent on land use system and farming practice.

AMF communities of grasslands colonized the plant roots faster than the AMF communities of the tilled arable lands in the trap cultures, while spore formation was significantly slower. These findings are in accordance with those obtained for grasslands and arable lands from Loess soils (Oehl et al., 2003a). However, through combined analyses of initial mycorrhizal root colonization and spore density in trap cultures, the AMF communities of either grasslands alone or arable lands alone could not be differentiated over a broad range of different soil types. Thus, first through analyses of the AMF species compositions, especially from field soil samples, differences between AMF communities across different soil types and land use systems could be revealed.

The effect of land use intensity on the AMF species communities dominated over the soil type factor (Fig. 6). Our results show that land use intensity effect is management dependent: Only monoculturing and narrow cereal mono-crop rotations with periodical non-vegetated soils, like in the sites PS-a, TS-a, DG-a1, DG-a2 and AG-a, had drastically reduced AMF species richness and diversities compared to the grassland systems (Table 4). In contrast, grasslands and arable soils with crop rotation or crop mixtures had similar

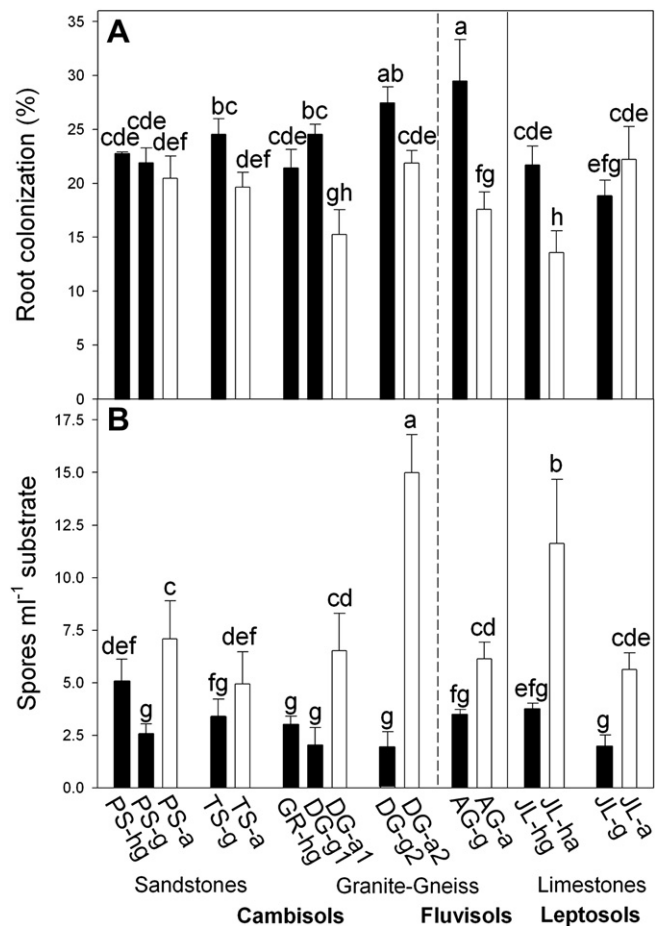


Fig. 8. Mycorrhizal root colonization (A) after 2 months, and AMF spore formation (B) after 4 months of trapping AMF in pot cultures; pots had been inoculated with soil derived from the field sites (see Table 1). Data are reported as averages and standard deviations for four replicate plots per site. Nonsignificant differences between sites are indicated by identical letters above the bars and were assigned by comparison with (Fisher's) Least Significant Difference at the 5% level after one-way ANOVA. Black bars represent grassland sites, while white bars represent arable lands. For site codes see Table 1.

AMF species diversities. This indicates that the presence of several plant species has a positive effect on AMF diversity. However, grasslands had much higher plant species diversity than arable fields with crop rotation or crop mixtures (Table 1), suggesting that above a certain basic level of plant/crop diversity, AMF diversity is not increasing with plant species diversity which is confirmed by the fact that for the grasslands plant and AMF species richness were not correlated. This is not in agreement with Burrows and Pfleger (2002) who observed an increase in AMF species richness with increasing plant species diversity. Johnson et al. (1992) hypothesized that soil factors and plant species may be of equal importance for the diversity of AMF species communities. However, our data show that soil characteristics (here soil type and pH) are much more important in regulating AMF species composition and in determining the occurrence of specific AMF species than plant species diversity.

This study revealed that many AMF species have preferred habitats and suggests that several species have specific and rather narrow niches. It would be interesting to know if for instance *Acaulospora*, *Scutellospora*, *Cetranspora* and *Gigaspora* species, preferably inhabiting the siliceous grasslands, fulfil the same ecological functions as a number of *Glomus* species that are characteristic for the calcareous grasslands. Most probably, these

species groups do not just replace each other in different soils since non-*Glomus* species usually are clearly seasonal (e.g. Pringle and Bever, 2002; Oehl et al., 2009) while *Glomus* species often show rather successional than seasonal patterns (Oehl et al., 2009). Moreover, the amount of extraradical hyphae formed by members of Gigasporaceae, and Scutellosporaceae and Racocetraceae (Oehl et al., 2008), are usually higher compared to those of Glomeraceae (Hart and Reader, 2002; Maherali and Klironomos, 2007). As a consequence it has been proposed that the contribution of the Gigasporaceae (and related families) to plant nutrient uptake is higher (Maherali and Klironomos, 2007). Moreover, the number of hyphal fusions and anastomosis is higher in Glomeraceae compared to Gigasporaceae (De la Providencia et al., 2005; Giovannetti et al., 2001, 2004; Avio et al. 2006) which suggests that *Glomus* species have a better adaptation to soil disturbance. Importantly, in high-input arable soils, the loss of specific species respective species groups might lead not only to a loss of genetic but also of functional AMF diversity, and thus, might cause a decrease of soil fertility and quality. It will be interesting to understand whether the loss of AMF diversity or the absence of specific niche species may represent a severe loss of ecosystem function. In this respect it is also important to know how fast 'specialist' species re-appear after land extensification or whether 'specialist' species can successfully be re-introduced.

In this study we used spore identification to determine AMF community composition. Molecular markers can also be used to characterize AMF communities and this has opened new perspectives for understanding AMF community dynamics in the field (e.g. Öpik et al. 2006, Hempel et al. 2007). However, due to sample size restrictions, until now, almost all studies using molecular tools focused on relative few field sites (Helgason et al., 1998; Daniell et al., 2001; Hijri et al., 2006; Šýkorová et al., 2007a) or investigated roots of co-occurring plant species at one site (Vandenkoornhuyse et al., 2002; Scheublin et al., 2004; Šýkorová et al., 2007b). Moreover, many of the primers used to characterize AM fungi do not detect all AMF taxa or are unspecific and amplify also non-glomeromycotan fungi (e.g. Alguacil et al., 2008; Krüger et al., 2009). Multiple primer sets, such as proposed by Redecker (2000) can also be used to cover a wide range of AM fungi. However, the use of multiple primers is more labor intensive, thus limiting the number of samples that can be analyzed. Hence, for comparisons of many field samples, such as in this study morphological identification has been the most appropriate tool for AMF community analysis. Ideally, one should perform both methods at least on a set of samples and doing so each technique could be calibrated.

5. Conclusions

We initially hypothesized that AMF communities are affected by soil types and that the differences might depend on the land use intensity in all soils. Indeed, we found that different soil types may have very different AMF communities and thus, can be characterized by their AMF communities. However, land use intensity had a greater impact on the AMF populations and AMF species composition than soil type and has to be taken into consideration when soil types are discussed and compared for their AMF populations. In our study, several AMF species and species combinations were characteristic for specific soil types and land use intensities. Vice-versa, we assume that within limits of error the prevailing AMF species can be predicted at specific land uses and soils. This also implies that AM fungi could serve as indicators to assess biological soil quality and fertility if the soil type is adequately considered. Future studies should explore how the differences in AMF species compositions in different soils and the

loss of specific AMF groups affect the soil properties and important ecosystem processes.

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